

Problem Statement ID: 26180

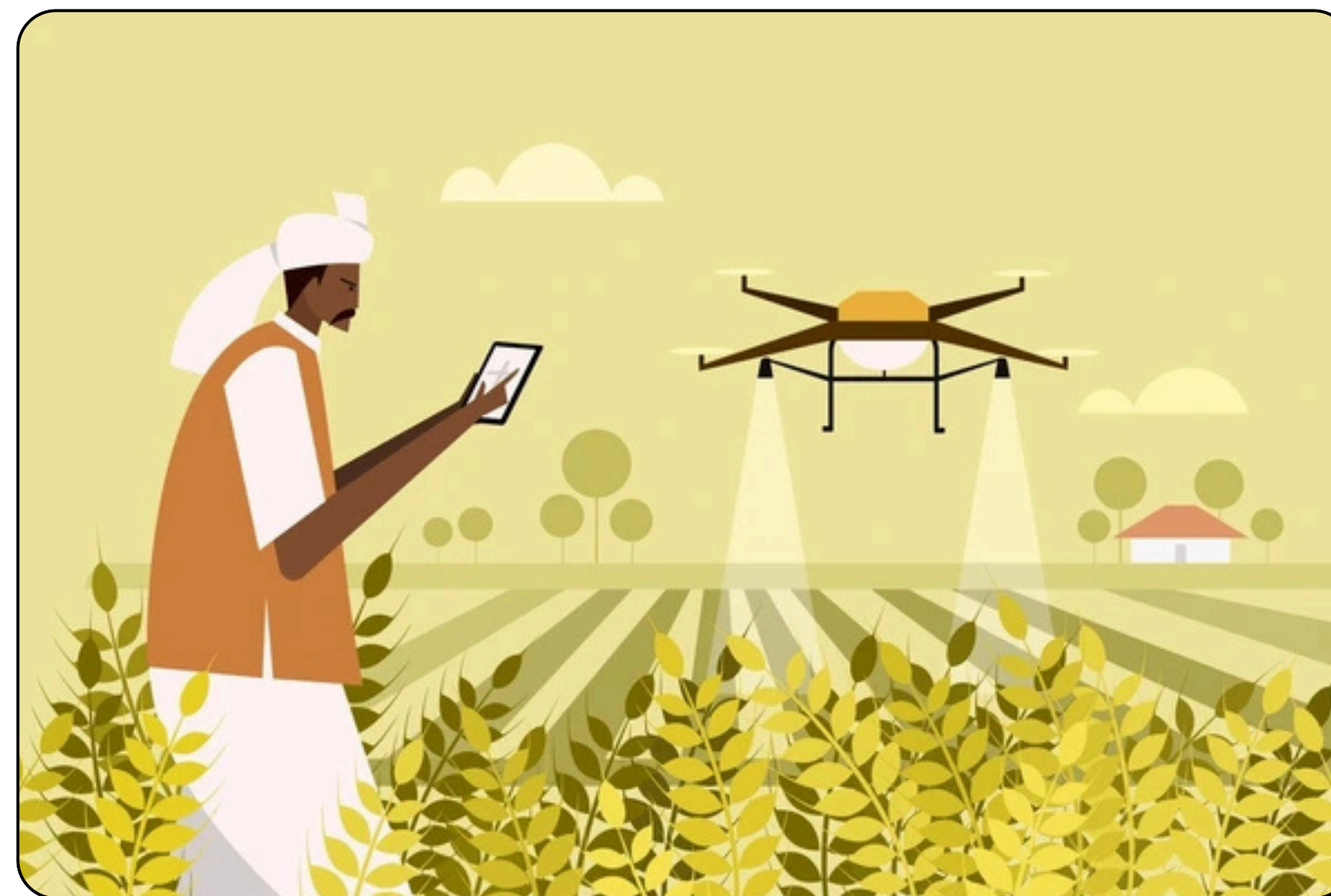
Problem Statement Title: AI-powered Smart Farming Assistant for real-time crop health, pest, nutrient, irrigation, and climate-risk detection

Theme: Agriculture, FoodTech & Rural Development

PS Category: Hardware

Team ID: H043

Team Name: Drikr





THE PROBLEM



Irregular Weather & Extreme Temperatures

- Unpredictable rainfall
- Heat stress on crops



Increasing Pest Attacks & Crop Failures

- Faster pest proliferation
- 20–40% crop losses



Rising Water Scarcity & Soil Degradation

- Depleting water resources
- Deteriorating soil health



High Costs & Limited Access

- Expensive agri-tech solutions
- Limited access for small and marginal farmers



THE SOLUTION



Real-Time Monitoring

- Track soil, weather & crop health
- Field sensors



Instant Alerts

- Detect abnormalities early
- Notify farmers on mobile app



Autonomous Response

- Drone inspects affected area
- Precision spraying as scheduled



Smart Resource Use

- Targeted intervention
- Less chemicals & water waste



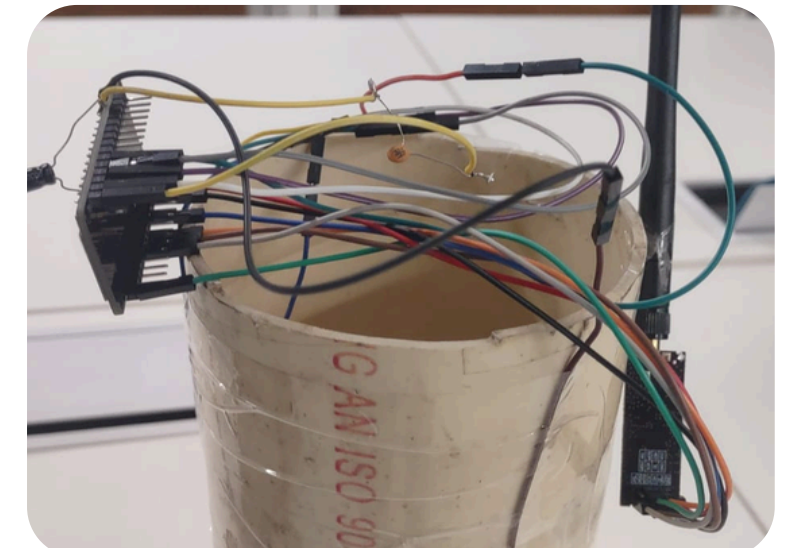
SENSORS IN THE FIELD
Continuous monitoring for healthier crops



INSTANT ALERTS ON MOBILE APP
Critical information in your hands



AUTONOMOUS ACTION
Precision spraying for healthier yields



HIGHER YIELDS



LOWER INPUT COSTS



EMPOWERED FARMERS



A SUSTAINABLE TOMORROW

System Workflow & Data Pipeline

End-to-end flow from field to farmer

1 Data Acquisition

Collect real-time data from multiple sources



2 AI Processing & Analysis

Turn data into actionable insights



3 Output Delivery

Deliver insights and actions to the field



Model Improvement

A continuously learning and evolving system





TECHNICAL FEASIBILITY

Built using proven, off-the-shelf and open-source technologies



IoT Sensor Network

- Soil moisture, pH, EC, NPK
- Temperature, humidity, VOC (BME688)
- Leaf wetness, light, rain, wind



Disease Biosensing

- Electrochemical biosensor
- Pathogen-specific reagents
- SPCE + potentiostat



AI/ML Models

- Multimodal analysis (sensor + image)
- Disease / pest detection & risk scoring



Drone Integration

- Aerial inspection (RGB/multispectral)
- Precision spraying (autonomous)
- Assisted scouting



Mobile Application

- Real-time alerts & recommendations
- Multilingual (Hindi, English + regional)
- Voice support



ECONOMIC VIABILITY

Affordable, scalable and sustainable

Prototype Development Cost (BOM)

Component	Approx. Cost (₹)
 Sensors (nodes, 4 units)	~ 44,000
 Biosensing module	~ 20,000
 Gateway & communication	~ 19,000
 Drone + spraying system	~ 40,000
 Mechanical, power & miscellaneous	~ 31,000
Total Prototype Cost	~ ₹1.54 Lakh
(Optional multispectral camera: + ₹15,000)	

Farmer Pricing Model (No large upfront cost)

 App + sensor monitoring	₹299 – 499 / month
Drone spraying service	₹400 – 600 / acre

Revenue Model (per year example)

 SaaS subscriptions (1,000 farms)	~ ₹47.9 L
Drone service margin (10,000 acres)	~ ₹10.0 L
Total Revenue	~ ₹57.9 L
Estimated Operating Profit	~ ₹18.9 L (≈33%)



OPERATIONAL FEASIBILITY

Practical, scalable and farmer-centric



24/7 Monitoring

- Continuous data collection
- Real-time alerts



Easy for Farmers

- Simple, actionable advice
- Spray, irrigate, monitor)



Multilingual Support

- Hindi, English and regional languages (voice + text)



Works in Low-Connectivity Areas

- Offline data storage
- Auto-sync when online



Deployment through FPOs

- Leverage existing agri-networks
- For large-scale adoption



Continuous Improvement

- Farmer feedback used to refine AI models and recommendations



FARMER IMPACT & COST REDUCTION

Lower input cost, higher yield, sustainable farming

Parameter	Conventional	With DRIKR	Reduction
Spraying cost (chemicals)	₹800 – 1,500	₹400 – 600	↓ 40–60%
Water usage	200 – 250 L	10 – 15 L	↓ ~90%
Labour cost	₹300 – 500	₹50 – 100	↓ 70–80%
Crop loss (pest/disease)	20 – 40%	5 – 10%	↓ 50–70%
Total cost per acre	₹1,100 – 2,000	₹500 – 700	↓ 30–60%



Potential saving for farmers:

₹400 – 1,300 per acre (30–60%)



KEY CHALLENGES AND MITIGATION

Challenge / Risk	Mitigation Strategy
 Sensor failure or drift	Regular calibration + redundant sensors
 Poor network connectivity	Offline storage + auto sync
 AI false alerts	Confidence thresholds + farmer confirmation
 Drone regulations & weather limits	Scheduled & supervised operations, weather checks
 Farmer adoption & trust	Simple UI, multilingual voice assistant, on-field training and support



₹1.54 Lakh
Prototype Cost



₹299 – 499 / month
Farmer Subscription



₹400 – 600 / acre
Drone Service



₹400 – 1,300 / acre
Potential Farmer Saving
(30–60%)



1. FARMER IMPACT

Empowered farmers, better decisions



Early disease & pest detection
Prevent spread before major crop damage



Plot-level recommendations
Right action at the right time



24/7 field monitoring
Less dependence on manual inspection



Multilingual voice assistant
Accessible to small & marginal farmers



More informed & self-reliant farmers



2. ECONOMIC BENEFITS

Lower cost, higher yield, greater income



30–60% reduction
in intervention cost



₹400 – 1,300 per acre
potential saving



Lower spending on chemicals, water & labour



Better yield protection
higher farm profitability



AI-based cost & market insights
support better decisions



Higher income
for a stronger tomorrow



3. ENVIRONMENTAL BENEFITS

Greener fields, cleaner future



~90% less water
for targeted spraying



40–60% lower chemical use
through precision intervention



Reduced unnecessary spraying
less pollution



Better soil & crop health
and higher biodiversity



More climate-resilient farming
for long-term food security



Sustainable agriculture
for future generations



4. SOCIAL & RURAL IMPACT

Stronger communities, inclusive growth



Hindi + regional-language access
for wider reach



Voice-first interface
for low digital literacy users



Farmer-to-farmer knowledge sharing
learn from peers and experts



FPO-based deployment
enables community-scale adoption



Supports small & marginal farmers
and reduces rural-urban divide



Empowered rural communities
and inclusive growth

RESEARCH AND REFERENCES

Ref. No.	Author(s)	Title	Methodology	What it Gives You	Limitation
[1]	Neupane & Baysal-Gurel (2021)	Automatic Identification and Monitoring of Plant Diseases Using UAVs: A Review	 UAV-based surveys	✓ UAV-based disease monitoring, survey role	! Review only; no closed loop to actuation
[2]	Mohanty, Hughes & Salathé (2016)	Using Deep Learning for Image-Based Plant Disease Detection	 54,306 images 14 species, 26 diseases (CNN)	✓ 99.35% accuracy (on held-out test set)	! Controlled conditions (lab, not field)
[3]	Singh et al. (2020)	PlantDoc: A Dataset for Visual Plant Disease Detection	 2,598 in-the-wild images 13 species (fine-tuning)	✓ Up to 31% accuracy improvement	! Internet-scraped; annotation not pathologist-supervised
[4]	Moupojou et al. (2023)	FieldPlant	 25,170 plantation images 8,629 annotated leaves (27 classes)	✓ Real-world plantation dataset with expert labels	! Limited crop range; small vs PlantVillage
[5]	Kharim et al. (2019)	Droplet deposition density at low-altitude UAV aerial spraying in rice	 Variable spray rate & flight speed analysis	✓ Deposition ↑ with spray rate, ↓ with speed (less drift at low speed, high rate)	! Single crop; fixed nozzle set
[6]	FAO / IPPC (2021)	Scientific Review on the Impact of Climate Change on Plant Pests	 Policy review	✓ Plant pests & diseases cut 20–40% of global yields per year	! Policy-level; no implementation guidance